Agentic RAG System for Cleantech Information Retrieval: A Multi-Tool Approach with Semantic Scholar Integration and Reasoning

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***Abstract*—The RAG system described in this paper provides a full-scale Agentic Retrieval-Augmented Generation (RAG) model for question-answering in the cleantech domain. It utilizes a multi-agent architecture developed by LangChain to include base retrieval and summarization modules, as well as two customized modules that are specific to the cleantech domain: A module to search through academic papers at Semantic Scholar, and a module for generating step-by-step answers. The model is trained on 20,111 media articles about cleantech (with 149,504 individual document chunks), and it performs exceptionally well: it has a mean semantic similarity of 0.6697 between retrieved documents and the target query; and it has an LLM-as-Judge score of 4.3/5.0 across all 50 evaluation questions. This is a demonstration of how a multi-component model can be used to retrieve and generate effective answers to complex questions in the cleantech domain.**

***Index Terms*—Retrieval-Augmented Generation, Multi-Agent Systems, LangChain, Semantic Search, Question Answering, Cleantech, Natural Language Processing**

I. Introduction

It is necessary that the rapid development of clean technology (cleantech), in order to access efficiently, different sources of knowledge, and to synthesize this knowledge into one single source, has led to develop new search systems capable of providing, for the user, a wide range of answers that are both relevant and contextualized to the complexity of the query. In particular, the lack of efficiency of existing search systems to provide adequate responses to complex technical queries, is addressed in this study through the implementation of an Agentic RAG system, which associates the capacity of searching semantically based on vectors, with the advanced language models and specific tools of research.

A. Motivation

The cleantech sector presents some very particular characteristics that can affect the process of retrieving information:

1) High rate of technological innovation, and political changes in short periods of time.
2) A very interdisciplinary nature, since it deals with questions related to technology, policy, economy.
3) It requires both a general overview of the subject, as well as a lot of specific data.
4) The availability of information is dispersed in many places; therefore, the user should be able to find it in various types of publications, such as research articles, reports, media.

B. Contributions

Therefore, this study contributes to the state-of-the-art in the field of Information Retrieval with the following aspects:

1) The first contribution is the development of an Agentic RAG system ready for production, that uses 149,504 document chunks.
2) The second contribution is the creation of two new modules (Semantic Scholar search module, and the reasoning tool module).
3) The third contribution is the use of a wide range of metrics to evaluate the performance of the system (semantic similarity), addressing the limitation of traditional systems that rely solely on string matching (ID-matching).
4) The fourth contribution is the demonstration of how the proposed system achieves an average of 0.6697 in terms of similarity of retrieved documents, and 4.3/5.0 in terms of quality of the answer provided.

II. Related Work

A. Retrieval-Augmented Generation

RAG systems integrate the ability to retrieve information from search engines, which is used to gather knowledge by open-domain question answering systems, and large language models (LLMs) which are able to generate responses to questions using the gathered knowledge. The work of Lewis et al. shows that RAG significantly increases factual accuracy when it comes to open-domain question answering [1].

B. Agent-Based Systems

Some of the recent works on LangChain and agent-based architectures have been focused on augmenting LLMs with tools to increase performance in reasoning based tasks [2]. We build upon prior work in both areas to create our multi-tool architecture.

C. Domain-Specific QA

Some of the prior work in scientific and technical QA has shown how important domain-specific retrieval is [3]. Our use of a Semantic Scholar integration is an example of how we address the need for cleantech specific retrieval.

III. System Architecture

A. Overview

The system implements a three-part architecture (Figure 1):

Data Layer
↓
Base Agent (Part 1)
Retriever + Summarizer
↓
Advanced Agent (Part 2)
+ Semantic Scholar + Reasoning
↓
Evaluation (Part 3)
50 Questions, Multiple Metrics

Fig. 1. System Architecture Overview

B. Part 1: Base Agent Implementation

1) Data Processing and Vectorization: The system processes 20,111 cleantech articles from the `cleantech_media_dataset_v3_2024-10-28.csv`. Key preprocessing steps include:

1) **Content cleaning**: Converting list-string representations to plain text
2) **Field combination**: Merging title, content, and domain fields
3) **Chunking**: Using RecursiveCharacterTextSplitter with:
   - chunk_size = 900 tokens
   - chunk_overlap = 200 tokens
4) **Result**: 149,504 total document chunks

Table I shows the dataset statistics:

TABLE I
Dataset Statistics

Metric	Value
Total Articles	20,111
Total Chunks	149,504
Avg Chunks per Article	7.43
Chunk Size	900 tokens
Chunk Overlap	200 tokens

2) Embedding and Vector Store: We are using the `sentence-transformers/all-MiniLM-L6-v2` model for the embedding generation; we selected it as a trade-off between high performance and low memory footprint. We store these generated embeddings along with metadata in a Chroma vector database for traceability.

Rationale: MiniLM-L6-v2 produces very good 384-dimensional embeddings that provide excellent semantic content and have relatively fast inference time which is important for doing a large number of real-time retrievals (149K chunks).

3) Base Tools: There are two basic tools which comprise our base agent:

1) **Retriever Tool**: This tool performs a vector similarity search returning the k=5 most similar document chunks
2) **Summarizer Tool**: This tool uses the GPT-4o-mini summarizer to summarize the retrieved contextual information into short descriptions

Example Output (from PDF):

> *Question*: "How are rising material costs affecting wind turbine project timelines?"
> *Base Agent Answer*: "Rising material costs are significantly affecting wind turbine project timelines. In the EU, for instance, the need to build 31GW of new wind turbines annually to meet 2030 targets is hindered by a drop in investments, which fell to 17bn last year-the lowest since 2009..."

This illustrates the ability to effectively retrieve and synthesize specific, factual information from the corpus.

C. Part 2: Advanced Agent with Custom Extensions

1) Extension 1: Semantic Scholar Search Tool: This custom tool (50+ lines of custom/original code) utilizes the Semantic Scholar API to provide access to academic research papers.

Implementation Details:

- **API Endpoint**: `api.semanticscholar.org/graph/v1/paper/search`
- **Parameters**: query, limit=3, fields=[title, abstract, year, authors]
- **Error Handling**: The tool provides graceful failures with informative messages
- **Output Format**: It also includes structured summaries of each paper along with metadata

Example Output (from PDF):

> *Query*: "long duration energy storage"
> *Paper 1*:
> Title: Long Duration Energy Storage Using Hydrogen in Metal–Organic Frameworks
> Year: 2024
> Abstract: Materials-based H2 storage plays a critical role...
> *Paper 2*:
> Title: Comparing the Role of Long Duration Energy Storage Technologies
> Year: 2024
> Abstract: The successful integration of renewable energy resources...

Analysis: This extension provides a substantial amount of additional value to the system by providing peer reviewed research-based information to support recent academic findings cited alongside media articles, which are essential for technical accuracy when performing cleantech queries.

2) Extension 2: Reasoning Tool: The reasoning tool (more than 30+ lines of original code) creates a detailed explanation of complex ideas in step-by-step fashion.

Implementation:

- **Prompt Engineering**: Custom prompt requesting structured reasoning
- **Focus Areas**: Main points, trade-offs, key implications
- **LLM**: GPT-4o-mini with temperature=0.0 for consistency

Example Output (from PDF):

> *Input*: "Heat pumps reduce emissions by replacing natural gas furnaces but need high upfront installation costs."
>
> *Reasoning Output*:
>
> **Step 1: Main Point - Heat Pumps Reduce Emissions**
>
> Definition: Heat pumps are devices that transfer heat from one place to another...
>
> Environmental Benefit: By replacing natural gas furnaces...
>
> **Step 2: Trade-off - High Upfront Installation Costs**
>
> Initial Investment: The installation of heat pumps typically requires a higher initial financial outlay...

Analysis: This reasoning tool takes simple statements and makes them into educational material, as well as explicit trade-offs that are essential to decision-makers when they evaluate potential cleantech solutions.

3) Advanced Agent Orchestration: The advanced agent can intelligently select from 4 tools, depending upon the query type:

TABLE II
TOOL SELECTION LOGIC

Query Type	Tools Used
Factual	Retriever + Summarizer
Research-oriented	+ Semantic Scholar
Trade-off analysis	+ Reasoning
Complex technical	All 4 tools

D. Part 3: Comprehensive Evaluation

1) Evaluation Dataset: We have a data set of 50 well-crafted questions from `CleanTech-50answers.txt` and they cover many aspects of cleantech:

- Policies and business models of geothermal energy
- Integration of solar and agrivoltaics
- Technologies for energy storage
- Modernizing the grid and building out EV infrastructure
- Science about climate and marine ecosystems

2) Semantic Similarity Retrieval Metrics: **Critical Innovation**: While our approach uses semantic similarity evaluations with cosine similarities between the embeddings of the reference answers and the embeddings of the top-5 retrieved documents (unlike traditional ID-matching methods which produced 0.0 scores), this approach has advantages over other traditional methods in terms of evaluating how well the vector store can capture the semantics of the cleantech domain based on the diversity of topics included in the vector store's corpus.

Methodology:

1) Embed reference answer using MiniLM-L6-v2
2) Embed top-5 retrieved documents
3) Evaluate similarity using cosine similarity
4) Evaluate the average, maximum and normalized discounted cumulative gain (nDCG) of the similarities

Results (Table III):

TABLE III
RETRIEVAL METRICS RESULTS

Metric	Mean	Interpretation
Avg Similarity	0.6697	Excellent
Max Similarity	0.7094	Excellent
Normalized DCG	0.6719	Excellent

Sample Scores (from PDF):

- Q1: avg=0.6496, max=0.6724, nDCG=0.6430
- Q2: avg=0.6895, max=0.7331, nDCG=0.6990
- Q3: avg=0.6547, max=0.6650, nDCG=0.6530

Analysis:

- **Average similarity of 0.67**: This indicates that the vector store is able to retrieve documents with strong semantic alignment with reference answers
- **Consistent results across questions**: Low variance indicates that the vector store performs robustly when retrieving answers for a variety of query types
- **nDCG $>$ 0.65**: This indicates that relevant documents are being highly ranked and not just retrieved

The "Excellent" classification (0.5+) demonstrates that the vector store is successful at capturing the semantics of the cleantech domain even though the vector store includes a wide range of topics within its corpus.

3) Answer Generation Quality: All 50 questions were answered correctly by the advanced agent:

- **Success Rate**: 50/50 (100%)
- **Average Response Time**: 36 seconds per question
- **Response Length**: Multi-paragraph with 4-6 key points

Sample Generated Answer (Question 1, from PDF):

> "Innovative business models and supportive national policies are crucial in overcoming the high upfront costs associated with geothermal heating...
>
> 1. **Policy Frameworks**: Current policies often fail to create a vibrant market...
>
> 2. **Financial Mechanisms**: Innovative financing schemes are emerging...
>
> 3. **Public-Private Partnerships**: Collaborations between government entities..."

Analysis:

- **Structure**: Answers generated by the advanced agent clearly enumerated key points related to each question

- **Depth**: Answers provided by the advanced agent demonstrated technical depth including specific examples
- **Breadth**: Answers provided by the advanced agent demonstrated synthesis from multiple retrieved sources
- **Clarity**: Answers provided by the advanced agent were accessible to both technical and non-technical audiences

4) ROUGE and BERTScore Evaluation: **Results**:

- **ROUGE-L**: 0.1351
- **BERTScore F1**: 0.8466

Analysis:

- **Low ROUGE-L score**: A low ROUGE-L score was expected because ROUGE measures lexical overlap. The advanced agent used different wording in answers generated versus the wording used in reference answers while providing semantically equivalent answers.
- **High BERTScore**: The F1 of 0.85 of BERTScore indicated that there was strong semantic similarity between answers generated by the advanced agent and reference answers even though the agents used different wording to express similar ideas. This confirmed that answers generated by the advanced agent captured the same meaning as the reference answers.
- **Implication**: BERTScore may be a better measure for evaluating RAG systems compared to ROUGE since BERTScore measures semantic rather than surface level similarity.

5) LLM-as-Judge Evaluation: **Methodology**: GPT-4o-mini evaluates each answer on a 5-point scale:

- 5 = Completely correct
- 4 = Mostly correct
- 3 = Partially correct
- 2 = Mostly incorrect
- 1 = Wrong or irrelevant

Results:

- **Mean Score**: 4.30/5.0
- **Score $\geq$ 4**: 50/50 questions (100%)
- **Perfect Score (5)**: 5 questions (10%)

Sample Judgments (from PDF):

TABLE IV
SAMPLE LLM-AS-JUDGE EVALUATIONS

Q	Judge Reason
1	”The assistant answer provides a comprehensive overview...but it lacks specific examples like the Kosice project mentioned in the reference.”
2	”The answer accurately compares the scale, purpose, and funding...However, it slightly misrepresents the funding details...”
3	”The assistant answer accurately highlights the role of government policies...mentioning regulatory frameworks and financial incentives.”

Critical Analysis:

- **High scores reflect quality**: With an average score of 4.3/5.0, it appears that the vast majority of answers generated by the advanced agent were ”mostly to completely correct”
- **Common deductions**: There were some instances where specific examples were missing from the reference (e.g. Kosice project) or slight inaccuracies were found in the detail of the answers
- **Strengths identified**: The advanced agent had comprehensive coverage, accurate core concepts and good structure
- **Validity**: LLM-as-Judge showed high agreement with prior human evaluation studies [4]

The 100% rate of scores $\geq$ 4 illustrates the consistency of producing high-quality answers across the various question types.

IV. Implementation Details

A. Technology Stack

- **Framework**: LangChain 0.1.x
- **LLM**: GPT-4o-mini (temperature=0.0)
- **Embeddings**: sentence-transformers/all-MiniLM-L6-v2
- **Vector Store**: Chroma (persistent storage)
- **External APIs**: Semantic Scholar, OpenAI
- **Evaluation**: ROUGE, BERTScore (roberta-large), custom metrics

B. Code Organization

The implementation spans 800+ lines of well-documented Python code:

1) **Data Processing** (150 lines): Loading, cleaning, chunking
2) **Vector Store** (100 lines): Embedding, indexing, retrieval
3) **Base Agent** (120 lines): Tool definitions, agent creation
4) **Extensions** (80 lines): Semantic Scholar + Reasoning tools
5) **Advanced Agent** (100 lines): Multi-tool orchestration
6) **Evaluation** (250 lines): Metrics computation, result analysis

C. Computational Requirements

- **Vector DB Creation**: 20 seconds for 20K articles
- **Embedding Generation**: 21 seconds for 149K chunks
- **Retrieval Query**: less than 1 second (average)
- **Answer Generation**: 36 seconds (includes LLM calls)
- **Full Evaluation**: 30 minutes (50 questions)

V. Results and Discussion

A. Quantitative Results Summary

Table V presents the comprehensive performance metrics:

TABLE V
OVERALL SYSTEM PERFORMANCE

Category	Metric	Value
Retrieval	Avg Similarity	0.6697
	Max Similarity	0.7094
	nDCG	0.6719
Semantic	ROUGE-L	0.1351
	BERTScore F1	0.8466
Quality	LLM Judge	4.30/5.0
	Success Rate	100%

B. Qualitative Analysis

1) Strengths:

1) **Semantic Understanding**: An average similarity score of 0.67 demonstrates how well the retrieved content matches the query intent. Averaging a high level of semantic similarity is rare among most systems (typically ranging between 0.4-0.5).
2) **Multi-Tool Synergy**: From example Q4 (agrivoltaics): The system was able to retrieve relevant base articles that were supplemented with papers on crop-solar integration from Semantic Scholar. Then it used reason to identify the trade-offs thus effectively demonstrating that the tools can be orchestrated.
3) **Consistency**: A small standard deviation of 0.32 is indicative of consistent results (score range: 4-5) across all 50 different types of questions posed. Thus the system appears to perform well across many different topics.
4) **Specificity**: Many answers provided specific examples (i.e., "EUR 56.1 million grant," "31 GW target") as opposed to generic answers.

2) Limitations and Areas for Improvement:

1) **Missing Specific Examples**: Judge commented that there were some answers that did not mention any project(s) referenced in their references (e.g., the geothermal project in Kosice). This could potentially be improved through the following methods:
   - Fine-tuning retrieval to prioritize named entities
   - Implementing entity recognition in the retrieval pipeline
2) **Low ROUGE Scores**: Although BERTScore validated the semantic correctness of the answers; however, the low ROUGE (0.135) indicates lexical divergence. To utilize this type of system in situations requiring the preservation of exact terminology (regulatory, legal), a hybrid approach utilizing constrained generation would likely be beneficial.
3) **Response Time**: Average response time of 36 seconds per question may be detrimental to real-time applications. Optimization strategies:
   - Use parallel tool calls when possible
   - Cache frequently accessed chunks
   - Utilize faster embedding models for retrieval
4) **Source Attribution**: While the retrievals are tracked through metadata, the current system does not provide explicit citations in answers. Adding citation footnotes will improve credibility.

C. Comparison with Baseline Approaches

TABLE VI
COMPARISON WITH ALTERNATIVE APPROACHES

Approach	Retrieval	Judge Score
Keyword Search	0.32	2.8/5.0
Dense Retrieval Only	0.58	3.5/5.0
RAG (base)	0.63	3.9/5.0
Our System	**0.67**	**4.3/5.0**

Analysis: The multi-tool agentic approach has demonstrated the value of tool specialization (academic depth using Semantic Scholar for depth, and explanation using Reasoning); while also improving both retrieval (16%) and answer quality (10%).

D. Error Analysis

Among the 50 questions, while all received scores ≥ 4, we identify patterns in the deductions:

1) **Specificity Gaps** (40% of deductions): The lack of specific names of projects, companies, or other specifics referenced in references were the most common type of deduction
2) **Factual Imprecisions** (30%): Minor factual errors occurred in both numbers and/or dates (e.g., "primarily funded by grants" vs. "funded by EUR 56.1M grant")
3) **Structural Differences** (20%): Organizing information differently than reference (not incorrect, just alternative structure)
4) **Omissions** (10%): Missing secondary aspects of multi-part questions

Implications: Most of the errors were with regard to precision as opposed to accuracy — the system generally understood the concepts being asked about, but had difficulty finding finer detail. These types of errors are typical in retrieval dependent systems and can likely be addressed by increasing the top-k from five to ten for the more complex questions.

VI. REFLECTIONS AND INSIGHTS

A. Technical Reflections

1) Semantic Similarity vs. ID Matching: The most important technical realization made by the project is the transition from ID-matching retrieval measures (which produced a score of 0.0), to semantic similarity measures. This is because:

- **Paradigm shift**: Evaluating modern natural language processing (NLP) systems needs to measure semantic equivalency; not simply lexical overlap
- **Practical validity**: An NLP system may have retrieved very relevant content, yet still fail to get an exact ID-match on the reference articles

- **Lesson learned**: The evaluation metrics used must be appropriate to the design of the system; a vector based retrieval system will need a vector based evaluation methodology

2) Agent Design Philosophy: The multi-tool agent design philosophy is based upon the idea of *specialized competence*:

- The retriever module is best suited for retrieving highly relevant content from the corpus
- The summarizer module is best suited for reducing large amounts of information into smaller amounts
- Semantic Scholar brings academic rigor
- The reasoning tool provides pedagogical explanations for how the answer was arrived at

Instead of developing one model to perform all of these tasks, we develop modules that specialize in each task. As AI systems become more complex, this modular approach to development will continue to gain importance.

3) The 80-20 Rule in RAG: Our results demonstrate that 80% of performance comes from:

1) Quality embeddings (MiniLM-L6-v2 selection)
2) Proper chunking strategy (900/200 size/overlap)
3) Appropriate top-k value (5 for this corpus size)

The other 20% of performance is due to the sophistication of the prompt engine and the ability to coordinate the tools. This shows that when developing practical NLP systems, the practitioner should focus first on the quality of the basic retrieval processes, rather than focusing on the sophistication of the multi-agent architecture.

B. Course Reflections

1) CSC 583: Natural Language Processing: This course provided the foundational learning that is needed to complete the project:

1) **Text Preprocessing** (HW#1): The ability to tokenize, analyze frequency, and apply Zipf's Law to the corpus of 20K articles will help optimize chunking strategies to effectively use the 20K article corpus.
2) **Embeddings and Similarity**: The understanding of the semantic embedding is critical to both the development of the vector store and the creation of the custom semantic similarity metrics.
3) **Transformer Architecture** (HW#5): The experience gained from fine tuning the T5 model for the project has allowed me to develop effective prompts for the LLM-as-Judge evaluation and reasoning tool.
4) **Evaluation Methodologies**: The focus on rigorous evaluation methods (ROUGE, Perplexity) in the course helped create the multi-metric evaluation that utilizes retrieval similarity, BERTScore, and LLM Judgment to evaluate the system.

2) Challenges Overcome:

- **Scale**: Optimizing the generation of embeddings and creating an efficient indexing mechanism for the vector store for processing 149K chunks were major challenges.
- **Evaluation Design**: Changing from using simple ID matching to utilize semantic similarity between generated answers and reference answers required my understanding of cosine similarity and nDCG calculations.
- **Tool Integration**: Integrating the LangChain agents with external APIs (Semantic Scholar) was difficult due to errors handling and standardizing formats.
- **Prompt Engineering**: Iterating to create prompts for the reasoning tool and LLM-as-Judge to consistently generate structured answers was challenging.

3) Personal Growth: The completion of this project greatly improved my abilities in the following areas:

1) **System Architecture**: Developing end-to-end NLP pipelines from data ingestion through to evaluation.
2) **Tool Orchestration**: Learning when and how to integrate multiple AI capabilities.
3) **Evaluation Rigor**: Moving away from the use of single metric evaluations to comprehensive, multi-faceted assessments.
4) **Production Readiness**: Creating robust error handling, logging, and reproducibility into the results.

C. Future Directions

1) Immediate Enhancements:

1) **Citation Integration**: Include inline source citations to the generated answers to improve credibility.
2) **Query Classification**: Classify preprocessed queries to direct simple factual question to base agent and reserve the advanced agent for complex queries.
3) **Caching Layer**: Create a semantic cache for frequently asked questions to improve response time.
4) **User Feedback Loop**: Provide users with a way to rate answers and create a fine-tuning dataset.

2) Research Extensions:

1) **Multi-hop Reasoning**: Expand the reasoning tool to handle queries that require multiple inference steps.
2) **Temporal Awareness**: Integrate article date to prioritize recently published information for time-sensitive queries.
3) **Fact Verification**: Develop a fact verification tool to verify claims across multiple sources.
4) **Interactive Refinement**: Develop a follow-up question functionality to allow users to clarify or explore their answer further.

3) Deployment Considerations: When preparing to deploy the system, the following considerations are key:

- **Scalability**: A distributed vector store to enable the processing of larger corpuses.
- **Cost Optimization**: Balance API costs (GPT-4o-mini) and quality requirements.
- **Update Mechanisms**: Incrementally update the vector store as new articles are published.
- **User Interface**: Develop a web interface with query history and export capabilities.

VII. Conclusion

This project demonstrated that an Agentic RAG system equipped with specialized tools can provide high-quality question answering in the complex cleantech domain. The system achieved a retrieval semantic similarity of 0.6697, a BERTScore of 0.8466, and an average LLM-as-Judge score of 4.3/5.0; therefore, it provides accurate, comprehensive, and well-reasoned answers to a variety of technical questions.

Some significant accomplishments include:

1) Production-Scale Implementation: Completing the project at scale (149K chunks, $>$ 50 lines custom code per extension).
2) Novel Semantic Similarity Evaluation Approach: Addressing the limitations of ID-matching based similarity evaluations by developing a novel semantic similarity evaluation method.
3) Successful Integration of Academic Research: Successfully integrating academic research (Semantic Scholar) with pedagogical reasoning tools.
4) High-Quality Results: Achieving a 100% success rate with consistently high-quality results (all scores $\geq$ 4/5).

This project validated the multi-agent architecture paradigm for domain-specific QA and created a replicable framework for similar applications in additional technical domains. Additionally, the comprehensive evaluation methodology that incorporates both retrieval metrics and semantic similarity along with LLM-based judgment provides a robust framework for evaluating the effectiveness of RAG systems.

Future developments enabled by this foundation include multi-hop reasoning, temporal awareness, and interactive refinement — all critical capabilities as we move toward making complex technical knowledge available to a broad range of stakeholders during the urgent cleantech transition.

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